

PHATTHAYA, Thailand

WMO: 484610

Lat: 12.9233N

Lon: 100.8657E

Elev: 199

StdP: 14.59

Time Zone: 7.00 (E07)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	69.8	71.9	54.5	63.6	73.7	57.7	71.6	75.3	22.3	78.1	19.5	78.1	10.1	40	0.372

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	8.4	92.3	80.3	91.3	80.0	90.3	79.7	82.2	89.5	81.6	88.6	81.0	87.8	7.6	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.3	158.2	86.5	79.8	155.6	86.0	79.2	152.6	85.4	46.2	89.7	45.5	88.8	45.0	88.1	85.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
18.3	14.9	13.3	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			66.6	94.6	2.1	2.7	65.1	96.6	63.9	98.2	62.7	99.7	61.2	101.7
			58.3	83.3	2.7	0.9	56.3	83.9	54.7	84.4	53.2	84.9	51.2	85.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	82.8	79.9	81.8	83.5	85.5	85.4	84.6	83.8	83.7	82.6	81.8	81.5	79.9
	DBStd	2.90	2.55	1.99	2.25	2.28	2.24	1.95	1.92	1.68	1.90	1.80	2.41	3.31
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11985	926	891	1040	1065	1098	1038	1049	1044	978	986	944	926
	CDD65	6510	461	471	575	615	633	588	584	579	528	521	494	461
	CDH74	74133	4186	4907	6729	7844	8210	7433	7094	6953	5940	5381	5098	4358
	CDH80	27410	1050	1503	2543	3636	3838	3216	2767	2622	1951	1579	1522	1183
Wind	WSAvg	4.7	4.6	4.3	4.4	4.0	4.2	4.9	5.7	5.3	3.9	3.7	5.5	5.9
Precipitation	PrecAvg	46.7	0.8	0.7	2.1	2.8	5.4	5.7	4.3	4.2	8.9	9.1	2.2	0.4
	PrecMax	61.6	3.1	2.7	7.4	7.8	10.7	13.1	9.5	10.0	15.9	15.2	6.9	1.7
	PrecMin	35.0	0.0	0.0	0.1	0.2	1.3	0.8	0.7	1.7	4.7	3.5	0.0	0.0
	PrecStd	6.9	0.9	0.7	1.7	1.8	2.7	2.9	2.4	2.1	2.9	3.0	1.7	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	89.4	90.5	91.7	93.8	93.5	91.6	90.5	90.5	89.9	89.8	90.6	89.8
		MCWB	74.1	75.8	78.8	81.0	81.6	80.3	79.2	79.2	79.3	79.2	78.2	76.9
	2%	DB	87.4	88.7	90.3	92.4	92.1	90.3	89.3	89.0	88.6	88.4	88.8	88.1
		MCWB	75.4	76.7	79.1	80.7	81.0	79.9	78.9	79.0	79.1	79.0	78.1	76.9
	5%	DB	86.3	87.6	89.3	91.4	90.9	89.3	88.4	88.1	87.6	87.5	87.7	86.9
		MCWB	75.6	77.0	79.1	80.5	80.5	79.6	78.8	78.8	78.9	78.8	78.1	76.6
	10%	DB	85.1	86.5	88.3	90.4	89.7	88.4	87.5	87.2	86.6	86.4	86.5	85.7
		MCWB	75.3	76.9	78.8	80.2	80.2	79.4	78.6	78.6	78.6	78.6	77.4	76.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	79.5	80.4	81.8	83.0	83.3	81.9	81.0	81.2	81.5	81.5	81.5	80.7
		MCDB	85.3	87.2	89.1	91.3	90.8	88.6	87.1	87.3	87.0	87.3	87.1	86.7
	2%	WB	78.6	79.2	80.7	82.0	82.3	81.1	80.3	80.4	80.6	80.6	80.4	79.4
		MCDB	84.2	85.8	87.9	90.0	89.4	87.6	86.5	86.5	86.3	86.6	86.3	85.3
	5%	WB	77.9	78.8	80.0	81.5	81.7	80.6	79.8	79.9	79.9	79.9	79.6	78.5
		MCDB	83.3	85.2	86.9	89.0	88.5	87.2	86.0	85.9	85.5	85.5	85.1	84.3
	10%	WB	77.2	78.2	79.5	80.9	81.0	80.1	79.2	79.3	79.2	79.2	78.8	77.5
		MCDB	82.3	84.3	86.1	88.1	87.6	86.5	85.4	85.2	84.6	84.4	84.1	83.2
Mean Daily Temperature Range	5% DB	MDBR	10.5	10.0	9.2	9.3	8.4	7.7	7.1	7.3	7.9	9.1	9.5	10.3
		MCDBR	11.6	10.9	10.1	9.9	9.2	8.4	7.9	8.1	8.7	10.3	10.7	11.0
		MCWBR	5.5	4.3	3.5	3.4	3.5	3.4	3.3	3.3	3.7	4.6	4.9	5.5
	5% WB	MCDBR	9.1	9.2	9.3	9.4	9.0	8.3	7.6	7.8	8.5	9.6	9.7	9.8
		MCWBR	4.1	3.5	3.6	3.7	3.9	3.7	3.6	3.6	4.1	4.8	4.8	5.0
Clear-Sky Solar Irradiance	taub	0.532	0.585	0.596	0.540	0.476	0.455	0.452	0.453	0.465	0.540	0.501	0.486	
	taud	2.014	1.896	1.910	2.092	2.325	2.393	2.397	2.380	2.331	2.080	2.173	2.174	
	Ebn at Noon	245	237	238	250	262	265	266	270	268	246	252	254	
	Edh at Noon	55	63	64	53	41	38	38	39	41	52	46	46	
All-Sky Solar Radiation	RadAvg	1597	1737	1773	1871	1690	1612	1558	1589	1539	1505	1557	1557	
	RadStd	101	85	135	121	115	83	88	103	88	97	118	81	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (29 neighbors)	N/A	N/A	N/A	N/A	+0.36	+0.36	N/A	N/A	+130	+130

Nomenclature: See separate page